

Quantitative Molecular Neurogenomics 2025

Module 1

ME.440.825

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COURSE OVERVIEW

Today's Learning Objectives

- Review Course, Overview, Syllabus, Grading, and Expectations
- Confirm/fix setup of computational environment(s) for the course
- Understand expectations for participation
- Understand motivations for modern genomics analysis tools in Neuroscience research
- Exploration of VSCode, bash terminal, & basic python navigation

Overview

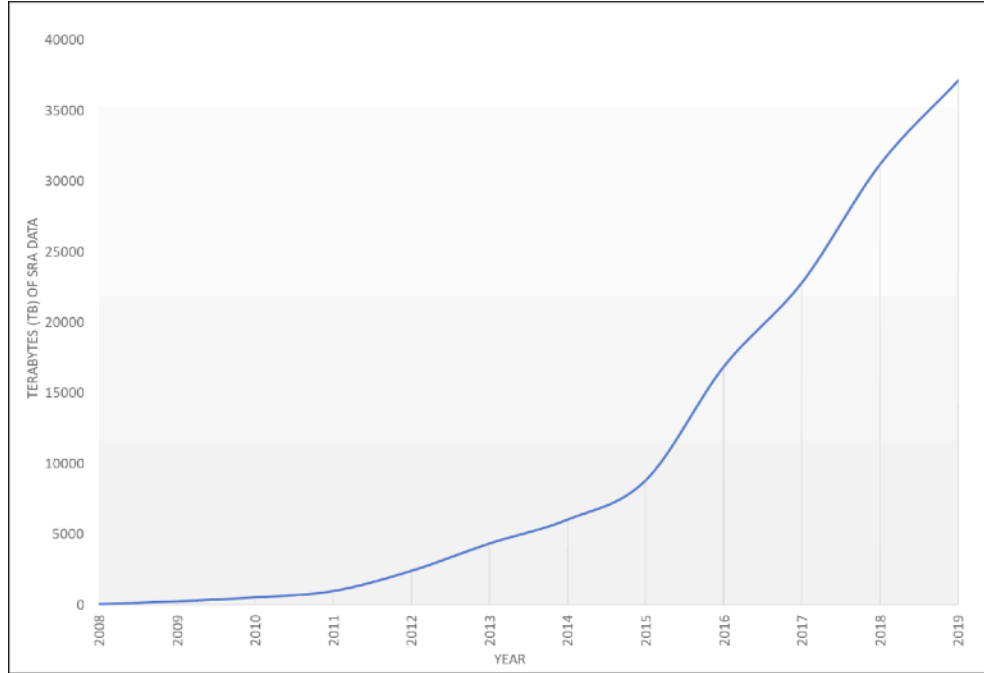
- Quantitative and computational methods are increasingly required for modern biological research.
- Neuroscience is no exception
- Through this course we want to empower you with the confidence and skills to apply these methods to high-throughput biological data.
- Where possible, we will highlight neuroscience-specific applications, resources, and methods



Why are we here?

- Modern biology is data intensive
- Biology is struggling to deal with the volume and complexity of sequencing data
 - Biologists need more training in data analysis
- Large-scale data is increasingly easier to generate and acquire
 - e.g., High-throughput sequencing & imaging
- But the complexity of experiments has grown as well
- Most modern biology **data** requires quantitative methods
 - Often methods are complex as well.

Growth of NIH SRA Database



Source: NCBI NLM Insights

Why can't I just outsource my data analysis?

- Biological data analysis is not a 'generalizable' problem.
- To **properly** analyze data, it's not enough to know how to use the tools. You also need to:
 - Understand the biological question & hypothesis
 - Understand the vagaries of the experimental assay & design
 - Have intuition about the correctness of the results
 - Recognize biologically relevant signals/patterns in the data
- ∴ Even in collaborations, you should understand and have a voice in how your data are being analyzed
 - Data analysis skills **must** be acquired by biologists
- “...if you're a biologist pursuing a hypothesis-driven biological problem, and you're using a sequencing-based assay to ask part of your question, generically expecting a bioinformatician in your sequencing core to analyze your data is like handing all your gels over to some guy in the basement who uses a ruler and a lightbox really well.” - Sean Eddy (<http://cryptogenomicon.org/>)



Course Learning Objectives

- Develop comfort working within a *NIX environment and with the command line to build and execute bioinformatics workflows
- Develop confidence in using python and other relevant frameworks and languages for bioinformatic analysis and visualization
- Develop knowledge of standard bioinformatic file formats and data structures and how they are connected in an analysis project
- Develop fluency in standard algorithms, statistical tests, and visualizations used in modern computational biology as applied to a diversity of applications.
- Develop good habits for reproducible research

Considerations

- This is the ~~first second~~ third year for this course
 - We are **still** learning how to teach this
 - Your feedback (in real time) will be appreciated
- The **primary objective** for this course is for you to learn and be exposed to principles of data analysis and visualization in genomics as applied to neuroscience
- We will **only scratch the surface** of modern genomics data analysis
- Your experience in this course will **largely** be self-guided, with help and resources made available as needed.
 - Lecture attendance alone is insufficient



Considerations

- There is going to be a high variance in skill set for this class
 - For those with less experience, you are in the right room
 - For those with more experience, you are in the right room
 - Collaboration and coordination across the class is *highly* encouraged.
- Please provide feedback often regarding the pacing of the course (too slow? Too fast?)



Course Goals

- Gain first-hand experience with various data analysis problems in euro-genomics
 - We will formally present only a fraction of problem types and workflows
- Become comfortable writing software and implementing available software tools to analyze large-scale biological data
- Develop skills and tools for reusable and reproducible data analyses
- Identify successful learning mechanisms, strategies, and resources to continue expanding your computational skills beyond this course

The Process

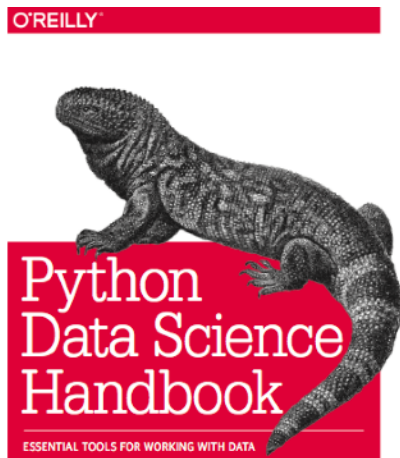
- ***Spend time*** with course materials, vignettes, publicly available tutorials, and published code to dissect individual components/steps in a workflow.
- Make mistakes and try different approaches.
- Find and reverse engineer existing code/analyses to understand how a tool or process is working
- Ask for guidance/support/help and help each other!
- This is just the beginning of a ***lifelong*** learning process
 - New workflows/tools are being developed every day
 - Data and experiments are getting more complex

Scripting is a foundational lab skill

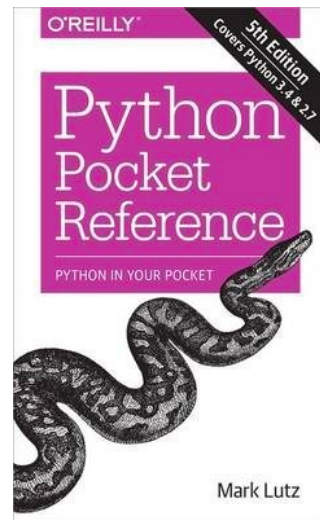
- Essential and easy to learn (like pipetting)
- Don't need to master a language to script
- Like learning to write a lab protocol
 - you learn by tweaking an existing protocol/script
 - Over time, you will better understand how to write your own from scratch
- Writing scripts and using command line tools may be intimidating but only because we are not yet familiar with these tools.

Recommended text

- Python Data Science Handbook
 - available for free at jakevdp.github.io/PythonDataScienceHandbook/
- Python Pocket Reference
 - Available for free at github.com/shahadot786/Python-Books/blob/master/python-pocket-reference-5th-edition.pdf



Jake VanderPlas



Google

- The *single greatest* asset to help your coding.
 - Search error messages (you're not the first to have this problem)
 - Find tool vignettes, walkthroughs, tutorials, and examples from which to build off.
 - Search for strategies/code to solve problems or roadblocks

GRADING

Grading

- Problem Sets (60%)
 - 10 problem sets, each worth 10 points
 - Lowest scoring pset will be dropped
 - Group/collective work is encouraged
 - Everyone must turn in ***their own*** version with name and date.
 - Due before next Monday class (11:30AM)
- Final Project (40%)
 - Re-analysis of public dataset / study of your choice
 - Reproduction of 2 original findings/figures.
 - 1 new question asked and addressed.
 - Choose a study that is interesting to you and/or relevant to your current research.
 - Outline of study and proposed analysis will be submitted and reviewed by TAs and course instructors (by week 4)
 - Final projects will be presented to class in the style of a brief lab meeting report (10 min)
 - Full analysis (repo, code, notebooks, etc) will be submitted via Canvas

Grading Rubric(s)

- Grading will be directly affected by effort and participation throughout the course.
- Problem set scoring is based on the demonstration of ***concerted*** effort, and not necessarily the correctness of results.
 - I.e. if you worked through the problems effectively but have a bug in your code that leads you to an incorrect answer, we will work with you to correct and not necessarily penalize.
- Final Project grade will be based on:
 - Effort (presentation & notebooks)
 - Ability to achieve analysis benchmarks established in project outline
 - Communication with TAs & Instructors throughout course

Final Project

- Learning the components of a complete *omics workflow is the primary goal for this course
- Start planning now
 - Identify paper/study/dataset early on
 - TAs and Instructors can help review selection
- Consider efforts made by original authors towards reproducibility in your selection
 - Well documented/detailed workflow
 - Code repository available?

Final Project

- Work **with** each other
 - Group work is encouraged in class and in research.
- Each of you will have a different path forward in your analyses
- Most of you will use tools/components in your workflow that we will not **explicitly** discuss in the course materials.
 - Use the learning/help strategies from this course to learn to connect tools for a proper workflow
- Check in **often** with instructors, TAs, classmates, and lab mates to help evaluate progress and troubleshoot.

Final Project - Deliverables

- Class presentation of study re-analysis
 - Study overview
 - Experimental Questions
 - Proposed re-analysis workflow
 - Comparison to originally published results
 - Presentation of novel finding/analysis
- Reproducible code base
 - Fully executed re-analysis
 - Repository (preferred) and/or compressed archive of all project-related analysis

Final Project - Resources

- Primary paper
- Project outline - submitted for review
- Vignettes and tutorials
- Instructors/TAs (please ask us questions. We are here for a reason.)
- Classmates
 - neurogenomics2025 Slack workgroup:
 - https://join.slack.com/t/neurogenomics2025/shared_invite/zt-3c242s338-fe~Fy_tJYVTVnBwo34PEyQ
- Package/tool/resource documentation
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Google Search

I'm Feeling Lucky

COMPUTE ENVIRONMENT(S)

Your compute needs will evolve during the course

- Local installation(s) at first
- As projects progress, you may need more:
 - Memory
 - Disk Space
 - CPUs
- Transition to a high-performance environment as needed
 - Rockfish (ARCH)
 - We will provide instructional material for accessing/operating w/in rockfish HPC

Rockfish

- Advanced Research Computing at Hopkins (ARCH)
 - State of the art high-performance computing resource
 - Allocation provided for this course
- Information:
 - Account Info & Resource Allocation: <https://coldfront.rockfish.jhu.edu/>
 - User Guide: <https://www.arch.jhu.edu/access/user-guide/>
 - FAQ: <https://www.arch.jhu.edu/access/faq/>

Check-ins

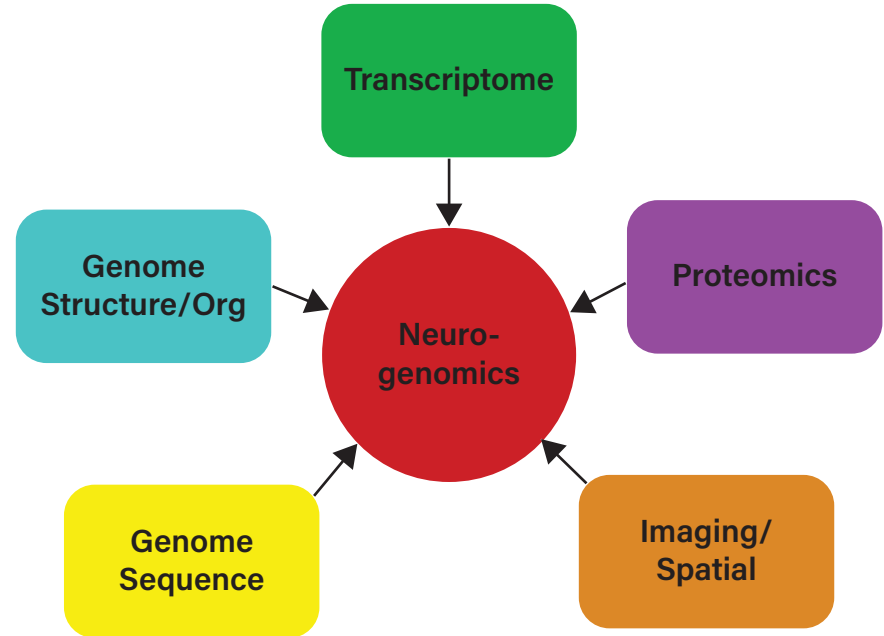
- Win/Mac/Linux?
- Prior computational/programming experience?
- Signed up for Rockfish access?
- Install VSCode locally?
- Complete/compile Module 1 Notebooks?
- Open and work on Module 1 pset?
- Experience with HPC?
- Experience with git/version control?

Questions about compute environment/setup?

MODERN [NEURO]GENOMICS: A BRIEF OVERVIEW

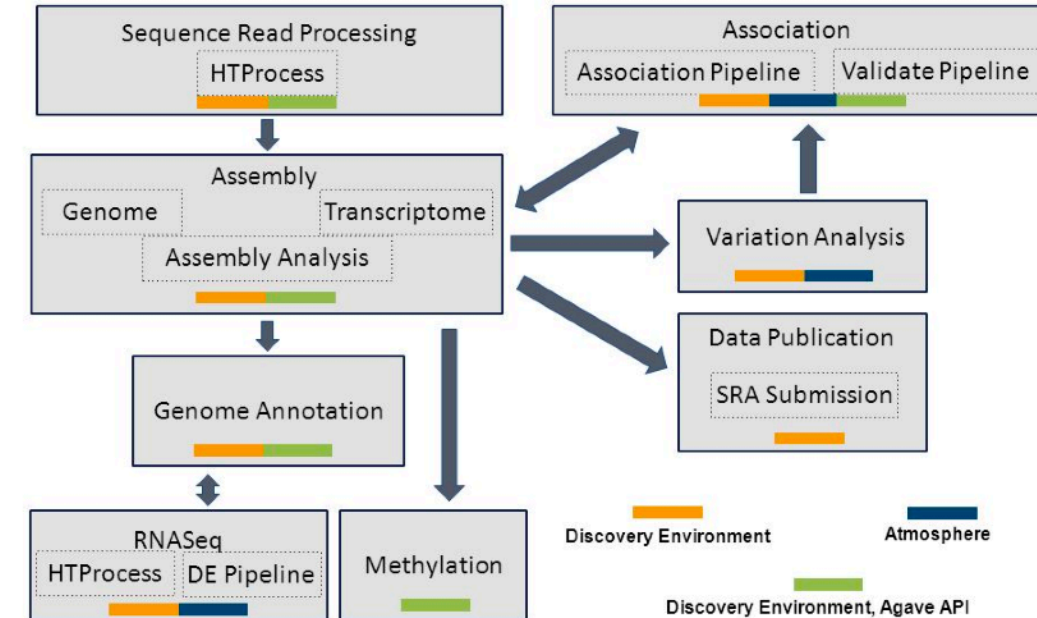
Neuro-genomics

- How does the genome of an organism influence the development and function of the nervous system?
- Genome and derivatives
- This course will focus on high-throughput sequencing-based assays as applied to neuroscience questions

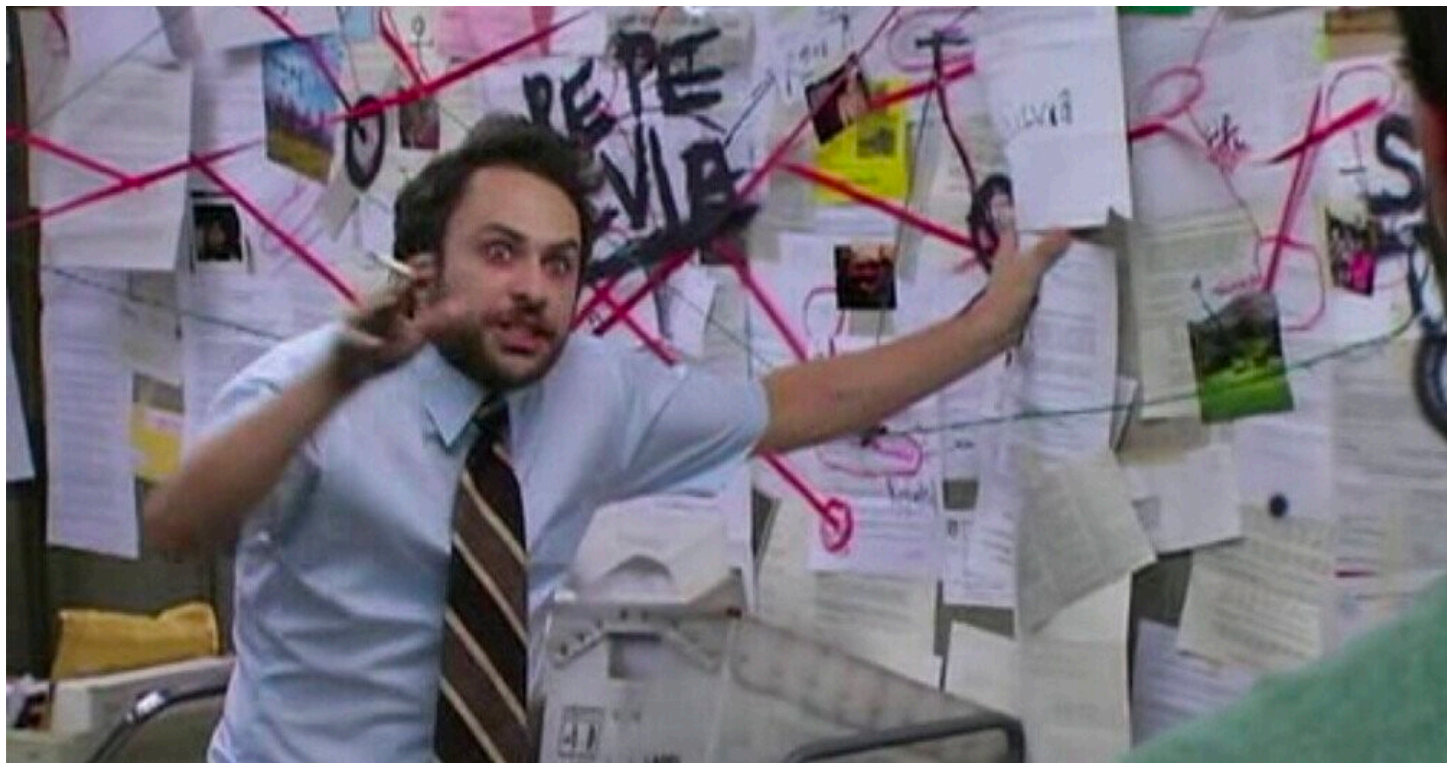


The theoretical workflow example

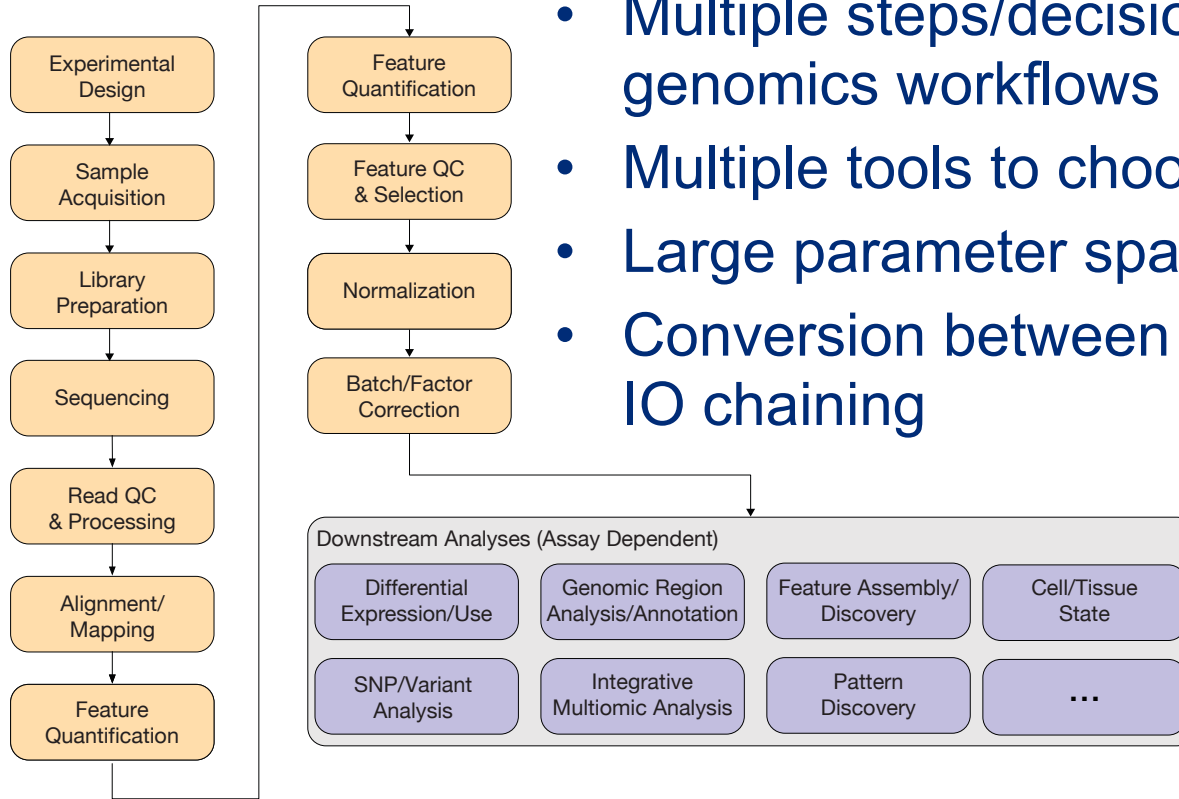
Overview of Genomics Workflows



The Reality

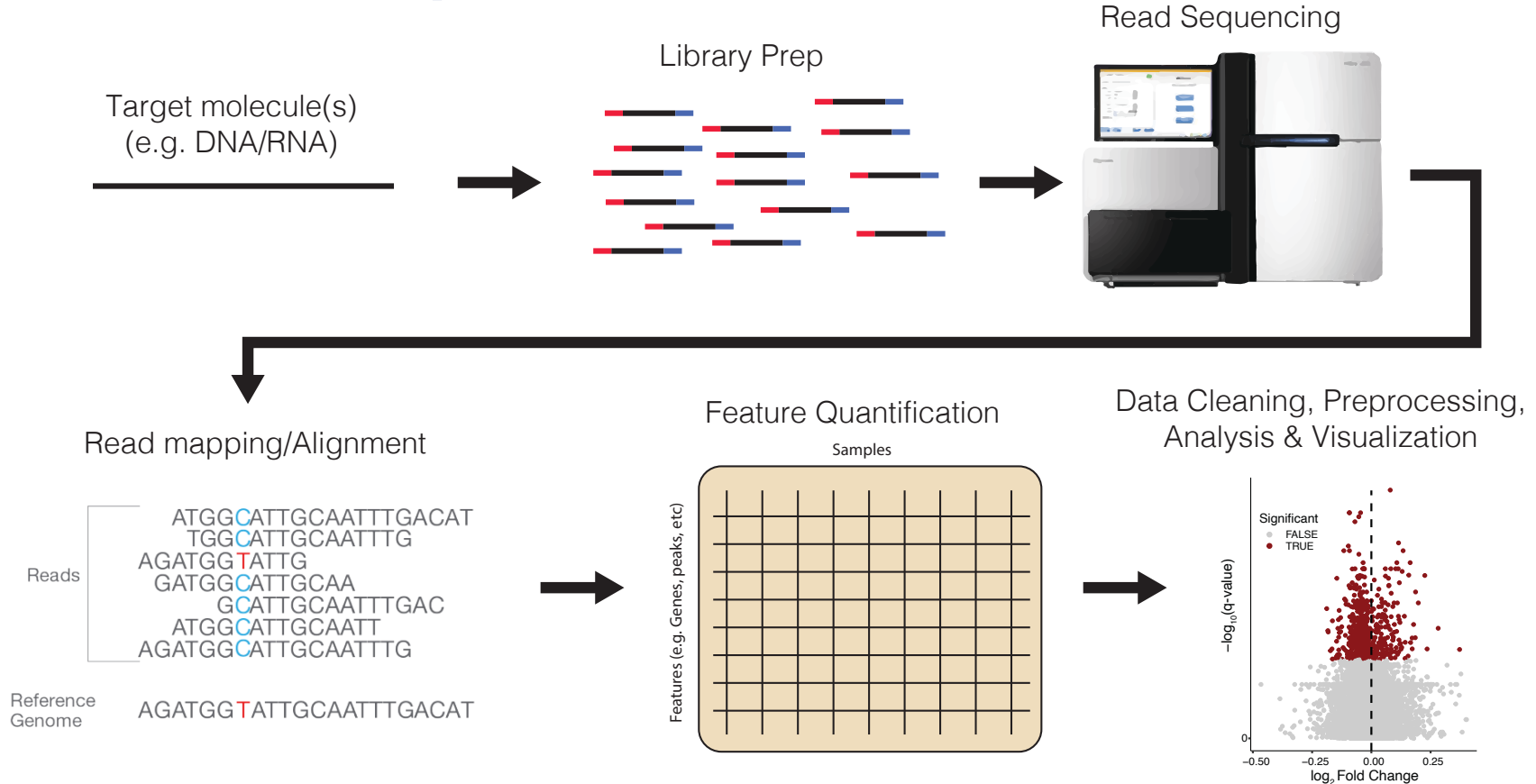


A (Grossly Oversimplified) Overview of *-omics Workflows



- Multiple steps/decisions in modern genomics workflows
- Multiple tools to choose from at each step
- Large parameter space for each tool
- Conversion between formats to facilitate IO chaining

Even more simplified



Sequencing is just a tool/assay

- Application still requires a hypothesis
 - There is *room* for discovery-based questions, but the strongest studies are still hypothesis-driven in their design and implementation.
- Many NGS assays are simply adaptations of ‘classical’ molecular biology assays to increase resolution and parallelization:
 - Diff RNA-Seq = high resolution, highly parallel Northern blot
 - ChIP-Seq = high resolution, highly parallel EMSA
 - Spatial Transcriptomics = highly parallel RNA in situ
- 100s of applications for nextgen sequencing of natural or exogenous nucleic acid sequences

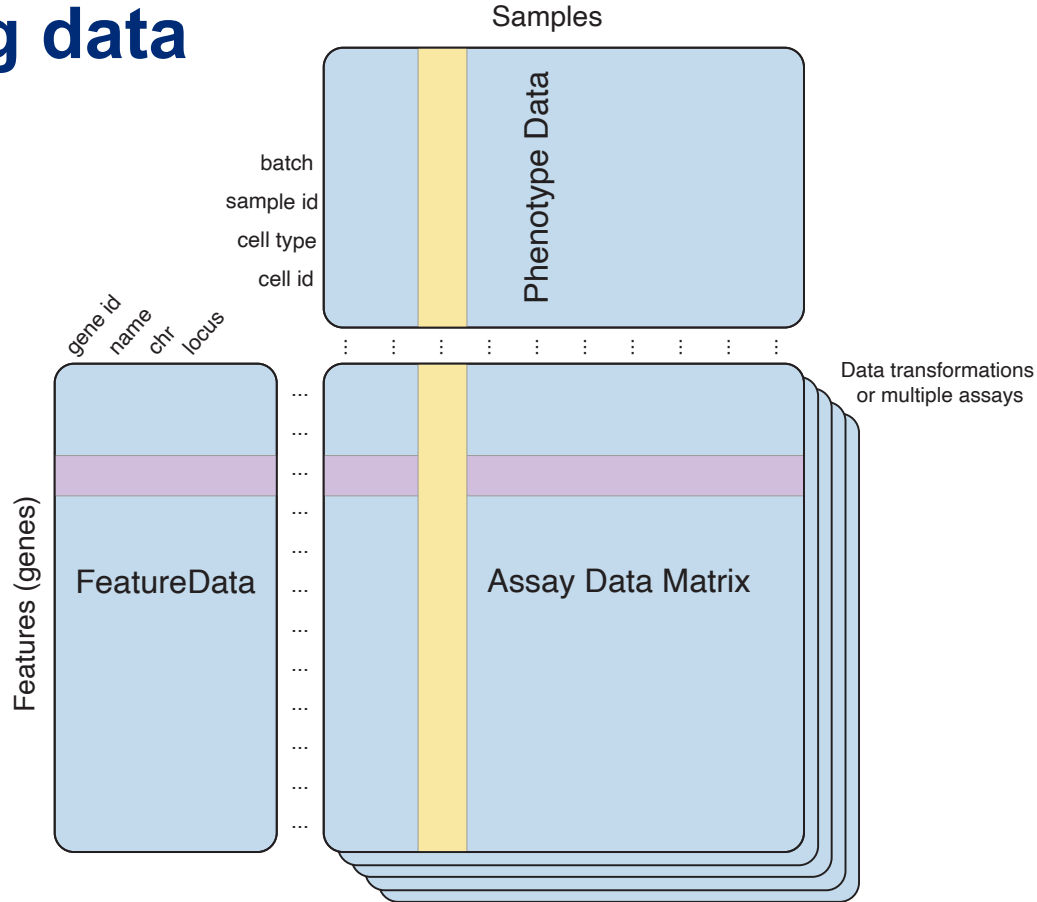
Computational Biology *is* Experimental Biology

- Do not ***always*** conflate high-throughput sequencing with ‘big data’-style science
 - Hypothesis-driven vs discovery science
- Important to define the question and understand the hypotheses to be tested
- Nature of high-throughput sequencing experiments allows for testing multiple hypotheses
 - e.g. differential analysis is equivalent to multiple independent pairwise tests between conditions (1 per feature)

NGS Sequencing Applications (Abridged)

- DNA
 - Whole Genome/Exome Sequencing
 - Targeted Sequencing
 - Hybridization Capture
 - Variant Identification
 - Gene expression profiling
 - Splicing & RNA dynamics
 - RNA modification analysis
- RNA
 - Whole Transcriptome Sequencing
- Epigenomics
 - ChIP-Seq
 - Methyl-Seq
 - Other DNA base modifications

A common data structure for high-throughput sequencing data

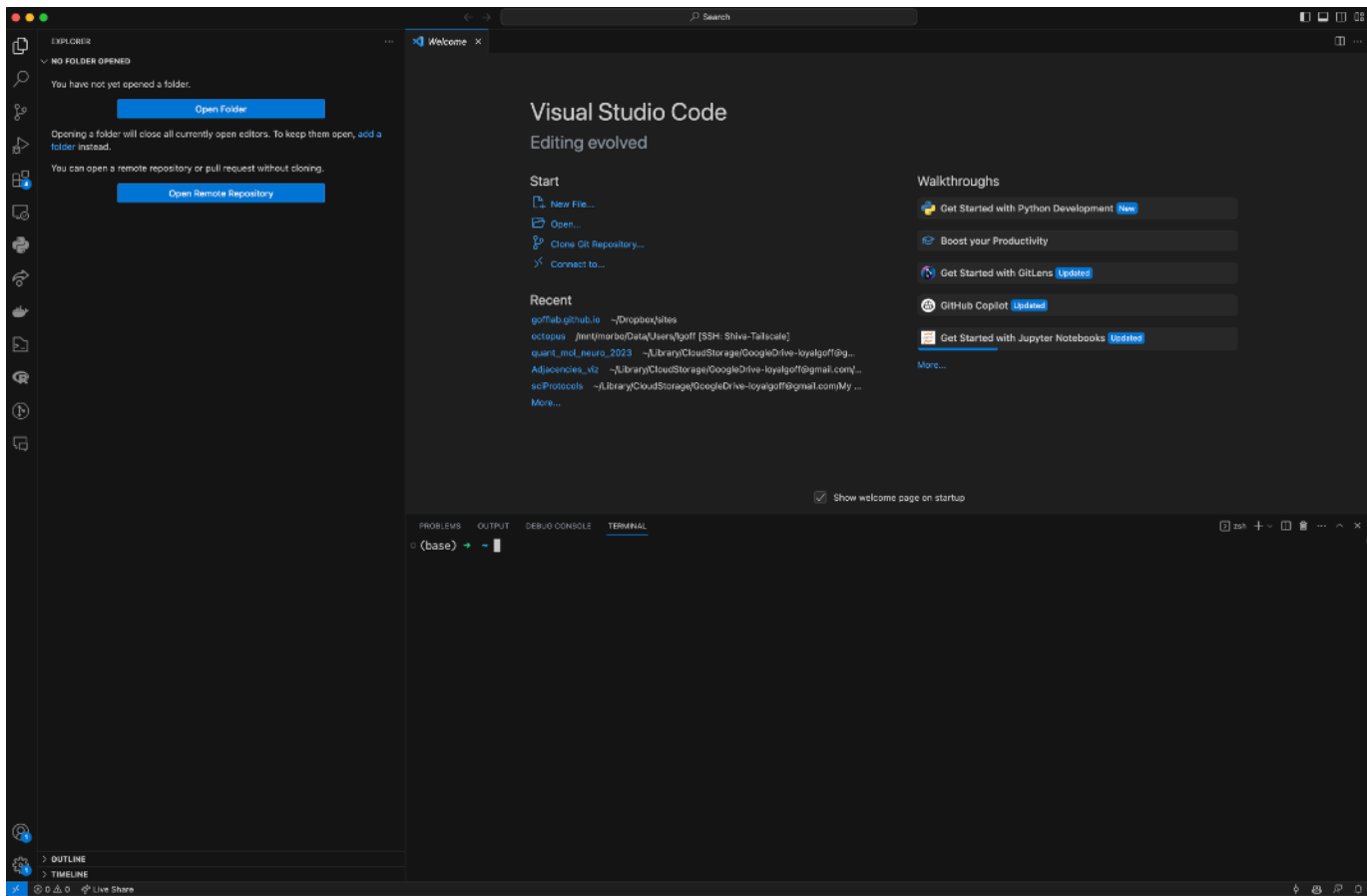


External resources

- <https://portlandpress.com/biochemist/article/43/6/58/229924/Beginner-s-guide-to-next-generation-sequencing>
- <https://www.illumina.com/areas-of-interest/neurogenomics.html>
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INTRO TO VSCODE, TERMINAL, AND PYTHON

VSCode - Integrated Development Environment (IDE)



PROBLEM SET 1

Questions?